

Jack Hogan

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Computer science and artificial intelligence engineer with experience across software engineering, AI/ML research, and technical leadership. Focused on deep reinforcement learning and systems that let models reason, adapt, and learn complex behaviors, with applications in robotics, vision, and low-data learning.

Experience

Era Labs

Embedded Systems (Software, Firmware, Electrical) Engineer

📅 August 2026 – Present 📍 San Francisco, CA

- Designed and implemented a patent-pending firmware architecture supporting dynamic peripheral registration and bytecode execution for ~100ms idea iteration.
- Designing and fabricating circuits for new product lines with microamp-scale power draws and efficient layouts.
- Designed a cohesive software stack for registering, finding, and collaborating with Beacons, a hardware/software solution for fostering real-time social connections at events, using traditional and LLM-based machine learning techniques.

Peraton Labs

Machine Learning Researcher

📅 May 2025 – May 2026 📍 Silver Spring, MD

- Created a new pipeline for concurrently testing different GNN architectures, speeding up testing by 2-3x.
- Investigated and implemented multiple bleeding-edge GNN architectures, improving task accuracy by 50%+.
- Developed and presented a technical report on a novel GNN architecture, demonstrating significant performance gains and guiding next-phase research.

Leidos Innovation Center

Software Engineering Intern, High-Performance Systems

📅 May 2024 – August 2024 📍 Arlington, VA

- Created and refined GPU-based signal processing algorithms in CUDA, Rust, and Python for real-time applications, improving performance by 400%.
- Developed a real-time machine learning system to lock onto hidden radar targets using PyTorch with 85%+ accuracy.

Peraton

Software Engineering Intern

📅 May 2023 – August 2023 📍 Chantilly, VA

- Automated document processing systems using AI-based parsing systems to increase processing efficiency.
- Authored and presented a whitepaper on systems and processes developed.
- Managed daily intern standups and kept tasks on track.
- Created data ingest and management dashboards for U.S. contractors.

Alluja LLC

Founder

📅 2022 – Present 📍 Chantilly, VA

- Creator of multiple products including mobile apps (SwiftUI), libraries (Swift, Rust), backend servers (Rust, Python), and frontends (Svelte, Leptos) impacting 5,000+ users.

Objective

Seeking work in deep reinforcement learning and reasoning: finding new ways to make models learn complex policies, with applications in problem analysis, critical thinking, robotics, vision, and low-data learning.

Education

Purdue University Honors College

Computer Science B.S., Artificial Intelligence B.S.; Minor in Mathematics; Graduated Spring 2026
📅 Fall 2022 – Spring 2026 📍 West Lafayette, IN

- Relevant Coursework: Data Structures, Graduate Machine Learning, Artificial Intelligence, Analysis of Algorithms, Abstract Algebra
- Study Abroad: University College Dublin, Spring 2024

Skills/Exposure

- | | | |
|--------------|-------------|-------------------|
| • Python | • PyTorch | • Actix Web |
| • Rust | • NumPy | • SwiftUI |
| • C | • SciPy | • Jetpack Compose |
| • C++ | • Pandas | • GStreamer |
| • C# | • CUDA | |
| • Java | • CUTLASS | |
| • Kotlin | • Burn | |
| • Swift | • SvelteKit | |
| • TypeScript | • Astro | |
| • JavaScript | • React | |
| • SQL | • Leptos | |

Tools

- EC2 • ECS • S3 • Docker • Docker Compose • MongoDB

- Komori, a Discord Siri Shortcuts integration, with paid downloads on the App Store.
- Alluja WebSockets, a WebSocket testing client for macOS, with 100+ downloads.

Research & Leadership

Purdue Hackers

Engineering Division Lead & Interim President

📅 Jan 2024 – May 2026

📍 West Lafayette, IN

- Oversaw all engineering projects for Purdue Hackers, a 2000+ member student organization focused on innovation and rapid prototyping.
- Managed a team of 10+ officers, balancing budget and time constraints while delivering multiple large-scale technical initiatives.
- Designed and maintained infrastructure for the Passports initiative, including authentication and management systems in Rust and TypeScript, supporting hundreds of active users.
- Led development of The Sign, a meter-tall Conway glider installation representing the club's logo; project was accepted to Hackaday and drew 1,000+ online views.
- Spearheaded the Beacons project in collaboration with Era Labs, deploying early prototypes at student events.
- Organized and ran major events, including a 200+ person callout, hack nights, and technical workshops.

Detecting Source Code Plagiarism in Submitted Assignments

Lead Researcher

📅 Spring 2023 – Fall 2025

📍 West Lafayette, IN

- Selected by Purdue faculty to lead an independent research project addressing plagiarism detection in the Computer Science program.
- Designed a language-agnostic detection system using Abstract Syntax Trees, parsing techniques, and machine learning to identify similarity beyond surface-level code patterns.
- Conducted large-scale testing across multiple courses, targeting 1,000+ student submissions annually to improve fairness and academic integrity.